ESP32 T12 / T15 soldering tip controller, 24 V — protected reference design

See docs/T12 Controller Design.md. Untested: prototype on a current-limited bench supply.

T5A
F1

Q4
PMOS −60V

J1
24V in

D5
12V

D2
SMBJ28A

C2
470u/35V

+24V_SW

R1
10m
HEAT_IN

R9
10k

Q3
PMOS −60V

R7
10k

D7
12V

R8
10k

BAT54S
D4

22k
R12

Q5
MMBT3904

C6
1u

R13
100k

C5
100n

HEARTBEAT

Q1
PMOS −60V

R5
10k

D6
12V

R6
10k

J2
T12 tip

4.7k
R10

PWM

Q6
MMBT3904

R11
100k

BAT54S
D1

10k
R4

OPA333
U3

C7
100n

100
R14

TC_ADC

C1
1n

C3
10n

249k
R2

R3
1k

The buck output also feeds the DevKit 3V3 pin: do not power the board from USB and 24 V at once.

LMR36510 values follow the datasheet 3.3 V example (22 uH, 3 x 22 uF out, 100k / 43.2k): check them.

C11
2.2u/100V

C12
220n/100V

LMR36510
U2

VIN
EN
VCC
BOOT
SW
FB
PGND
PAD
PG

100n
C13

22u
L1

FB

C14
1u

C15
22u

C16
22u

C17
22u

R17
100k

R18
43.2k

D3
3.6V

C9
10u

C10
100n

INA240A2
U4

C8
100n

+24V_SW

HEAT_IN

100
R15

CUR_ADC

C4
10n

R16
100k

NTC_NODE

J3
NTC handle

TH1
NTC 100k

ESP32-DevKitC
U1

EN
TC_ADC
NTC_NODE
CUR_ADC

CHIP_PU
32K_XP/GPIO32/ADC1_CH4
32K_XN/GPIO33/ADC1_CH5
MTDI/GPIO12/ADC2_CH5
MTCK/GPIO13/ADC2_CH4
MTMS/GPIO14/ADC2_CH6
MTDO/GPIO15/ADC2_CH3
VDET_1/GPIO34/ADC1_CH6
VDET_2/GPIO35/ADC1_CH7
SENSOR_VP/GPIO36/ADC1_CH0
SENSOR_VN/GPIO39/ADC1_CH3
5V
3V3
GND
U0TXD/GPIO1
U0RXD/GPIO3
GPIO0/BOOT/ADC2_CH1
ADC2_CH2/GPIO2
ADC2_CH0/GPIO4
GPIO5
GPIO18
GPIO19
GPIO21
GPIO22
GPIO23
DAC_1/ADC2_CH8/GPIO25
DAC_2/ADC2_CH9/GPIO26
ADC2_CH7/GPIO27
SD_CLK/GPIO6
SD_DATA0/GPIO7
SD_DATA1/GPIO8
SD_DATA2/GPIO9
SD_DATA3/GPIO10
CMD
GPIO16
GPIO17
UART_TX
UART_RX
LOOP_TIMING
PWM
HEARTBEAT

ESP32-DevKitC

LOOP_TIMING
J4
Loop pin

GND
UART_TX
UART_RX
EN
J5
Krumpir

Everything on +24V_SW must be rated 60 V or more: the TVS can clamp as high as 45 V.

Layout: one star ground at the negative terminal of C2. The heater return, the buck converter and the analog ground each run to it separately: keep R4, D1, C1, U3 and R2/R3 off the heater return.

R1 is a 2-pad 2512 shunt: route the INA240 inputs as a Kelvin connection at its pads.

+24V_SW
GND
+3V3

Sheet: /
File: t12-controller.kicad_sch

Title:

Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.6 | Id: 1/1